Bank Customer Churn Analysis Using Python and Power BI

Business Problem

The bank is experiencing customer churn and needs to understand which customer segments are more likely to leave. Analyzing historical customer data can help identify churn patterns and support data-driven customer retention strategies.

Dataset Description

The dataset selected for this project is a Bank Customer Churn Dataset. It contains customer information such as credit score, geographical location, age, account balance, banking product usage, account activity, and estimated salary. The dataset is used to analyze historical customer churn and identify customer segments associated with higher churn rates.

Attribute

Description

`RowNumber`

Sequential row number of each record.

`CustomerId`

Unique identification number assigned to each customer.

`Surname`

Customer's surname.

`CreditScore`

Credit score assigned to the customer.

`Geography`

Country where the customer is located.

`Gender`

Gender of the customer.

`Age`

Age of the customer in years.

`Tenure`

Number of years the customer has been with the bank.

`Balance`

Amount of money held in the customer's bank account.

`NumOfProducts`

Number of banking products used by the customer.

`HasCrCard`

Indicates whether the customer has a bank credit card (1 = Yes, 0 = No).

`IsActiveMember`

Indicates whether the customer is an active member (1 = Yes, 0 = No).

`EstimatedSalary`

Estimated annual salary of the customer.

`Exited`

Indicates whether the customer left the bank (1 = Yes, 0 = No).

`Total-Records`

10,000

Bank Customer Churn Analysis — Key Findings

Key Insight 1: **Overall Churn and Retention**

Total customers: 10,000

Churned customers: 2,037

Retained customers: 7,963

Overall churn rate: 20.37%

Overall retention rate: 79.63%

- Approximately one in five customers in the dataset left the bank, while about four in five remained.

- Although most customers were retained, the bank lost 2,037 customers. Understanding which customer groups have higher churn rates could help the bank identify areas for further investigation and improve retention strategies.

Key Insight 2: **Demographic Segments**

- Churn by age group

Age group	Total customers	Churned customers	Churn rate
18–25	611	46	7.53%
25–35	3,542	301	8.50%
35–50	4,586	1,127	24.57%
50–70	1,128	552	48.94%
70–100	133	11	8.27%

- Customers aged 50–70 have the highest churn rate at 48.94%. The 35–50 group also deserves attention because it has 1,127 churned customers, the largest number among the age groups.

- Older customers in this dataset show a higher historical churn rate, while the 35–50 age group accounts for the largest number of churned customers. The bank should investigate the needs and experiences of these groups rather than assume age itself causes churn.

- Churn by geography

Geography	Total customers	Churned customers	Churn rate
Germany	2,509	814	32.44%
Spain	2,477	413	16.67%
France	5,014	810	16.15%

- Germany has the highest churn rate at 32.44%, nearly twice the rate in Spain or France. Germany also has the highest number of churned customers, at 814.

- Germany is an important segment for further investigation. The bank could examine differences in customer experience, product suitability, or service issues in that market.

Key Insight 3: **Account Activity and Product Usage**

- Churn by account activity

Account status	Total customers	Churned customers	Churn rate
Inactive	4,849	1,302	26.85%
Active	5,151	735	14.27%

- Inactive customers have a churn rate of 26.85%, compared with 14.27% for active customers.

- Inactive customers are an important group for the bank to investigate. Potential actions include checking whether they are dissatisfied, having service difficulties, or no longer finding the bank's products useful.

- Churn by number of products

Number of products	Total customers	Churned customers	Churn rate
1	5,084	1,409	27.71%
2	4,590	348	7.58%
3	266	220	82.71%
4	60	60	100.00%

- Customers using three products have a very high churn rate of 82.71%. All 60 customers using four products are marked as churned in this dataset.

- The three- and four-product groups warrant investigation. However, the four-product group is small, and its 100% churn rate should not be generalized to all banking customers. The bank should verify the data and investigate whether the product combinations or customer circumstances explain the pattern.

- Churn by tenure

Tenure group	Total customers	Churned customers	Churn rate
0–2 years	2,496	528	21.15%
3–5 years	3,010	625	20.76%
6–8 years	3,020	570	18.87%
9–10 years	1,474	314	21.30%

- Churn rates are fairly similar across the tenure groups, ranging from 18.87% to 21.30%. The 9–10-year group has the highest rate, but the difference is small.

Key Insight 4: **Financial Characteristics**

- Churn by credit-score range

Credit-score range	Total customers	Churned customers	Churn rate
Poor	2,362	520	22.02%
Fair	3,331	685	20.56%
Good	2,428	452	18.62%
Very Good	1,224	252	20.59%
Excellent	655	128	19.54%

  - Customers in the Poor credit-score category have the highest churn rate at 22.02%, while the Good category has the lowest at 18.62%. The differences are relatively modest.

  - Credit-score range alone does not appear to separate customers into dramatically different churn-risk groups. The bank should examine it alongside other customer characteristics.
- Churn by account balance

Balance range	Total customers	Churned customers	Churn rate
200K–250,898	33	18	54.55%
1–50K	75	26	34.67%
100K–150K	3,830	987	25.77%
150K–200K	935	205	21.93%
50K–100K	1,509	300	19.88%
Zero balance	3,617	500	13.82%

  - Customers with balances above 200K have the highest churn rate at 54.55%, but this group contains only 33 customers. The 100K–150K group deserves greater attention because it has 3,830 customers and 987 churned customers.

- High-balance customers may warrant further investigation, but the small size of the highest-balance group means its churn rate should be interpreted cautiously. The bank should also investigate the 100K–150K segment because of its substantial contribution to customer churn.

- Churn by estimated salary

Salary range	Total customers	Churned customers	Churn rate
5K–10K	244	59	24.18%
100K–200K	5,010	1,044	20.84%
10K–50K	1,945	388	19.95%
50K–100K	2,537	504	19.87%
0–5K	264	42	15.91%

- The 5K–10K salary group has the highest churn rate at 24.18%, while the 100K–200K group has the largest number of churned customers, at 1,044.

- Estimated salary alone does not show a clear, consistent relationship with churn. The bank should investigate the 100K–200K group because of its size, while treating the 5K–10K result cautiously because it contains only 244 customers.

Key Insight 5: **High-Priority Customer Segments**

Customer segment	Customers	Churned	Churn rate
Germany, inactive, 1 product	720	375	52.08%
Spain, inactive, 1 product	548	178	32.48%
Germany, active, 1 product	629	203	32.27%
France, inactive, 1 product	1,253	371	29.61%

- The Germany–inactive–one-product segment is the strongest priority in this analysis. It has 720 customers, of whom 375 churned, giving it a churn rate of 52.08%.

- The bank should first investigate the Germany–inactive–one-product group, then assess the other high-churn segments. Potential retention actions should be designed only after the bank understands the reasons behind these patterns.

Conclusion

This project analyzed 10,000 banking customers and identified an overall churn rate of 20.37% using Python and Pandas. The analysis highlighted high-churn segments based on age, geography, account activity, product usage, and financial characteristics. These insights can help the bank prioritize customer segments for further investigation and develop targeted retention strategies. An interactive Power BI dashboard will present these findings to support data-driven decision-making.